

The End of the American Dream? Inequality and Segregation in US Cities

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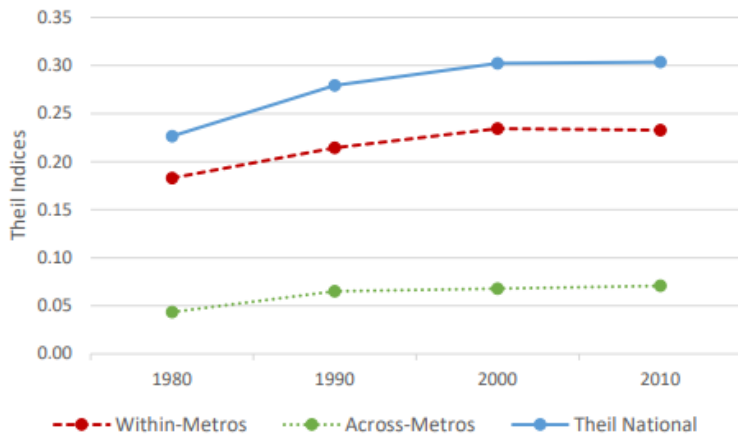
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Empirical Context (1980–2010):

- Simultaneous rise in US family income inequality and residential segregation.
- **Within-city** inequality drives national trends (much more than across-city).
- Strongest pattern observed specifically in **families with children**.

Motivation



Empirical Facts & Segregation Measure

Dissimilarity Index for Metro j :

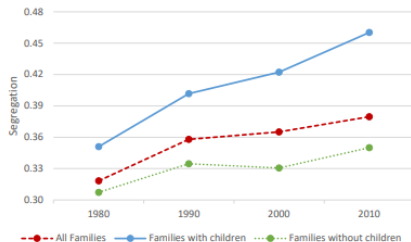
$$D(j) = \frac{1}{2} \sum_i \left| \frac{x_i(j)}{X(j)} - \frac{y_i(j)}{Y(j)} \right|$$

where $x_i(j), y_i(j)$ are poor/rich families in tract i , and $X(j), Y(j)$ are totals in metro j .

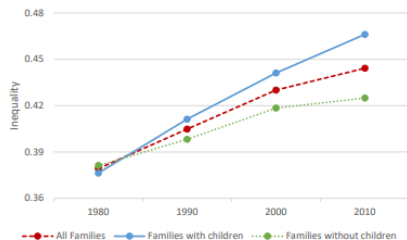
Key Empirical Findings:

- **Trend:** Gini (0.38 $\rightarrow$ 0.44) and Dissimilarity (0.32 $\rightarrow$ 0.38) rose jointly (1980–2010).
- **Cross-Section & Changes:** Initial inequality correlates with segregation; $\Delta \text{Inequality} \propto \Delta \text{Segregation}$.
- **Childhood Effect:** Segregation is markedly higher for households with kids $\implies$ points to **neighborhood exposure effects**.

Empirical Trends (1980–2010)



(a) Segregation



(b) Inequality

Simple Model: Ability & Local Spillovers

1. Child Ability Transmission

Log ability $\log(a_t)$ follows an AR(1) process across generations:

$$\log(a_t) = x + \rho \log(a_{t-1}) + v_t, \quad v_t \sim \mathcal{N}(0, \sigma_v^2)$$

2. Local Spillover Definition (S_{nt})

Spillover in neighborhood n equals the **expected future income** of children raised there:

$$S_{nt} = \frac{\int \int \int_{n_t(w_t, a_t)=n} w_{t+1}(w_t, a_t, \epsilon_t) F_t(w_t, a_t) \Psi_t(\epsilon_t) dw_t da_t d\epsilon_t}{\int \int_{n_t(w_t, a_t)=n} F_t(w_t, a_t) dw_t da_t}$$

- Captures school quality, peer effects, social networks, and role models.

Simple Model: Household Optimization & Wage Function

1. Child Wage Function w_{t+1}

$$w_{t+1} = \Omega(w_t, a_t, e_t, S_{nt}, \epsilon_t)$$

2. Parent Optimization Problem

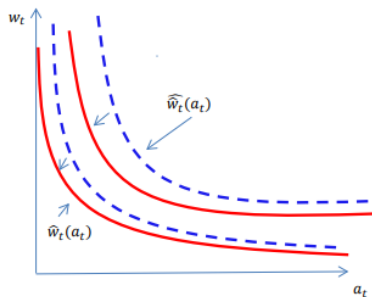
Parent with wage w_t and child ability a_t solves:

$$\begin{aligned} \max_{c_t, e_t, n_t} \quad & u(c_t) + \mathbb{E}[g(w_{t+1})] \\ \text{s.t.} \quad & c_t + R_{n_t}t + \tau e_t \leq w_t \end{aligned}$$

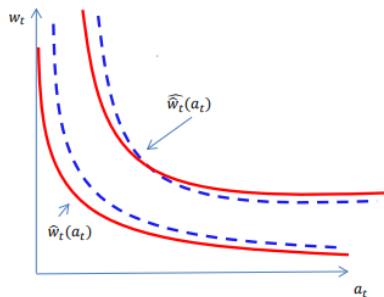
- High S_{nt} requires paying higher housing rent $R_{At} > R_{Bt} = 0$ [cite: 1].

Equilibrium Sorting & Skill Premium Shock

Figure 7: Cut-Off Response to Skill Premium Shock



(a) Partial Equilibrium



(b) General Equilibrium

General Quantitative Model Extensions

To map the model to data, the paper adds **four key extensions**:

- 1 3 Neighborhoods** ($n \in \{A, B, C\}$): Ordered $S_{At} > S_{Bt} > S_{Ct}$ and $R_{At} > R_{Bt} > R_{Ct}$.
- 2 Continuous Education** ($e \in \mathbb{R}_+$): Cost function τe^γ .
- 3 Preference / Amenity Shocks** (θ_n, ζ_n): Incorporates local amenities and idiosyncratic noise.
- 4 Endogenous Housing Supply:**

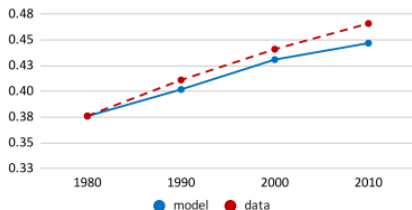
$$H_{nt} = \lambda_n \left(\frac{R_{nt}}{\bar{w}_t} \right)^{\phi_n} \quad (\phi_n = \text{housing supply elasticity})$$

Calibration Strategy (1980 Steady State)

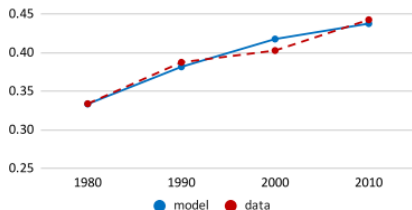
- **Baseline Year:** Calibrated to 1980 average US Metropolitan Area.
- **Macro Targets:** Skill premium (0.39), Gini (0.376), Dissimilarity index (0.334).
- **Disciplining the Spillover Strength:** Matched to Chetty & Hendren (2018b) micro-estimates:
 - 1 SD better county exposure $\Rightarrow$ **+6.2%** adult income (25th percentile parents).
 - 1 SD better county exposure $\Rightarrow$ **+4.6%** adult income (75th percentile parents).
- **Shock:** One-time, unexpected permanent rise in η to match the 1980–1990 rise in college premium.

Model Fit: Inequality & Segregation Dynamics

Figure 8: Responses to a Skill Premium Shock



(a) Inequality



(b) Segregation

Counterfactuals: How Much Does Segregation Matter?

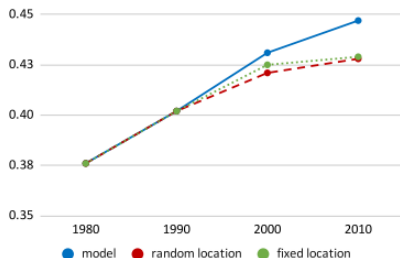
Counterfactual Exercises:

- 1 **Random Relocation:** Families are randomly reassigned across neighborhoods every period $\implies S_{At} = S_{Bt} = S_{Ct}$.
- 2 **Fixed Location:** Families cannot move away from their origin neighborhood after the shock.

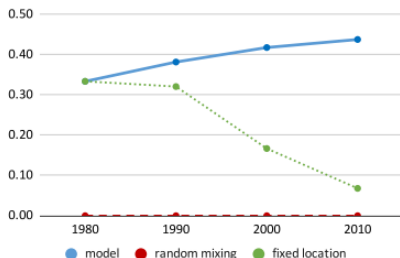
Main Quantitative Results

- Segregation accounts for **25%–27%** of the total increase in US income inequality between 1980 and 2010.
- Segregation accounts for **32%–33%** of the decline in intergenerational mobility (rise in rank-rank correlation).

Counterfactual Results: Muting Spatial Sorting



(a) Inequality



(b) Segregation

Policy Counterfactual: Scaling Up MTO

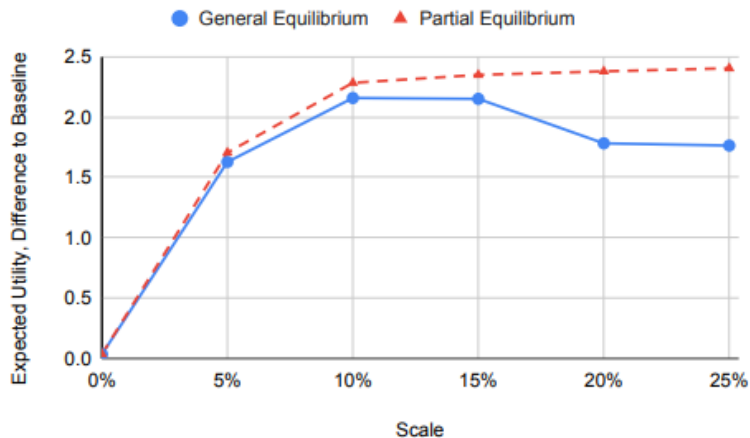
Moving to Opportunity (MTO) Vouchers:

- Lowest income families in neighborhood C receive vouchers to move to A .
- Financed via a city-wide income tax ν .

General Equilibrium Effects at Scale:

- **Small Scale** ($p \rightarrow 0$): +13% child adult income gain (matches Chetty et al. 2016).
- **Large Scale (Scaling up eligibility):**
 - 1 Inflow of poor families reduces spillover in A ($S_A \downarrow$) and improves C ($S_C \uparrow$).
 - 2 Higher demand drives up rents in A ($R_A \uparrow$).
- **Takeaway:** Child income gains and total social welfare follow an **inverted U-shape**.

MTO Scaling: GE Effects & Welfare



Conclusions & Policy Takeaways

- 1 Amplification Channel:** Residential income segregation amplifies skill premium shocks, accounting for over a quarter of the post-1980 rise in inequality.
- 2 Intergenerational Mobility Trap:** Rising rents price talented children from poorer families out of good neighborhoods, suppressing their educational return.
- 3 Limits to Scaling Vouchers:** Voucher programs (MTO) cannot be scaled linearly without triggering offsetting general equilibrium effects on rents and spillovers.